



GE MRI

THE PRESCAN PROCESS – HOW THE NMR SPECTRUM IS GENERATED

To generate that NMR spectrum peak on the screen during a pre-scan, the MRI machine strips away the spatial imaging gradients and treats the patient's body exactly like a chemistry lab spectrometer.

Behind the scenes, the scanner executes a precise sequence of five core physics steps to capture, process, and display the signal.

1

EXCITATION (THE RF PULSE)

The scanner transmits a non-selective, uniform RF pulse (usually a 90-degree pulse) through the body coil.

Because the spatial gradients are turned off, this pulse excites all protons across the entire volume at once. It knocks their net magnetization vector down into the transverse (xy) plane.



WHAT HAPPENS

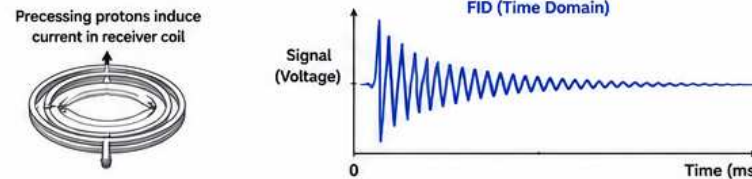
- All protons excited at once (no spatial selection)
- Net magnetization (M_0) tipped into the transverse (xy) plane

2

FREE INDUCTION DECAY (FID)

The moment the RF pulse shuts off, the protons begin to precess freely in the horizontal plane. As they rotate, their moving magnetic fields cut across the receiver coil loops. This induces an electrical current in the coil via Faraday's Law of Induction.

This raw, decaying electrical signal is called the Free Induction Decay (FID).



WHAT HAPPENS

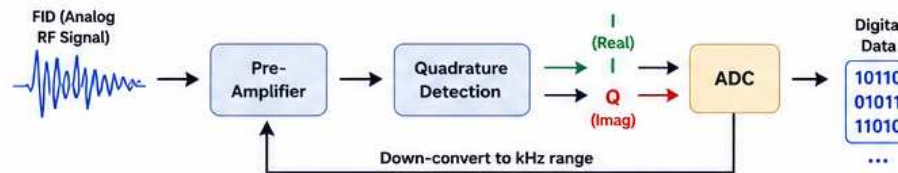
- RF off, protons precess freely
- Induced current in coil
- Raw decaying signal = FID

3

DIGITIZATION AND QUADRATURE DETECTION

The raw electrical signal is a complex, high-frequency analog wave (e.g., oscillating at 63.8 MHz for a 1.5T scanner). The scanner passes this signal through a pre-amplifier. It uses Quadrature Detection to split the signal into two channels (Real and Imaginary components) to determine the direction of rotation.

It down-converts the frequency to a manageable range (kilohertz) and digitizes it using an Analog-to-Digital Converter (ADC).



WHAT HAPPENS

- Amplify weak signal
- Split into I (Real) and Q (Imaginary)
- Down-convert frequency
- Digitize with ADC

4

THE FOURIER TRANSFORM (THE CORE MATH)

At this stage, the backend computer is holding a raw dataset of signal amplitude plotted over Time.

To turn this into a peak, the scanner runs a Fast Fourier Transform (FFT) algorithm.

The FFT converts the data from the Time Domain to the Frequency Domain.



WHAT HAPPENS

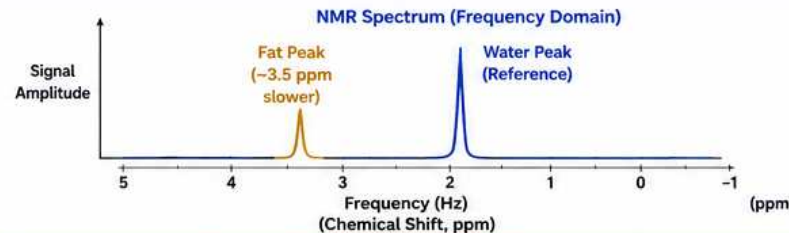
- FFT converts Time → Frequency
- Produces the NMR spectrum

5

GENERATING THE PEAKS

Because different chemical environments cause protons to spin at slightly different speeds (chemical shift), they show up at different frequencies on the resulting plot:

- **Water Protons:** Shielded less by electrons, they experience a slightly stronger field and spin faster.
- **Fat Protons:** Heavily shielded by dense carbon-hydrogen chains, they experience a weaker local field and spin slower (approximately 3.5 parts-per-million slower than water).



WHAT HAPPENS

- Chemical shift separates different proton signals
- FFT separates overlapping frequencies
- Plot: Frequency (X-axis) vs Signal Amplitude (Y-axis)
- Produces distinct water and fat peaks



KEY TAKEAWAY

By removing spatial gradients and using RF excitation, signal detection, digital processing, and the Fourier Transform, the MRI system becomes a sophisticated spectroscopy machine—producing the water and fat peaks you see during the pre-scan.

